

Dark Matter as a Trapped K=6 Remnant in the Octahedral Voids of the FCC Vacuum Lattice

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Abstract

Building on the Selection-Stitch Model (SSM) [2, 3], in which baryonic matter is identified with K=4 remnants trapped in the tetrahedral voids of the K=12 FCC vacuum lattice, we examine the framework’s second interstitial site — the octahedral void — as a candidate dark matter trap. The bonded subgraph of the octahedral defect is the complete tripartite graph $K_{2,2,2}$. Four qualitative predictions follow from its structural symmetry: absence of first-order electromagnetic coupling, absence of SU(3) color in the baryonic form, self-conjugate (Majorana-type) character, and suppressed first-order radiative cooling. The framework predicts the dark matter mass directly from a closed inclusion-exclusion expansion on $K_{2,2,2}$, terminating at third order because the octahedron’s six vertices forbid any 4-matching: $C_{\text{DM}} = 25 \cdot 144 - 30 \cdot 10 + 8 \cdot 8 = 3364$, giving $m_{\text{DM}} = (3364/1836) \times m_p = 1.719$ GeV. This prediction agrees with the recent Kang et al. 2026 detection of a 1.5–1.6 GeV gamma-ray line in three active galactic nuclei within 1–2.1 σ of each individual source measurement (joint fit at ~ 1.55 GeV, $\sim 11\%$ below the prediction). Since the line is observed in AGN, which host supermassive black holes and are predicted to develop dark-matter density spikes, gravitational redshift of inner-spike annihilation provides a candidate astrophysical mechanism mapping the rest-frame prediction to the observed line position.

1 Introduction

The cosmological abundance of dark matter relative to baryonic matter is one of the most precisely measured ratios in physics:

$$\frac{\Omega_{\text{DM}}}{\Omega_{\text{b}}} = 5.36 \pm 0.04 \quad (\text{Planck 2018 [1]}). \quad (1)$$

No mechanism in the Standard Model produces this number; it is an empirical input to current cosmological models. Candidate dark-matter frameworks typically introduce one or more new particle species with adjustable couplings and abundances, fit to the observed ratio, and test the result against direct-detection, indirect-detection, and structure-formation constraints.

The Selection-Stitch Model (SSM) [2, 3] proposes that the physical vacuum is a Face-Centered Cubic (FCC) crystallization of spacetime, with baryonic matter identified as a single K=4 node trapped in the tetrahedral interstitial void of the K=12 FCC bulk. The framework derives the proton-to-electron mass ratio

$$\frac{m_p}{m_e} = (K + 1)K^2 - c_{\text{skew}} K = 13 \times 144 - 3 \times 12 = 1836 \quad (2)$$

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from purely structural counts of the trapped tetrahedral-void defect [2]. An equivalent derivation via a [[192, 130, 3]] CSS code on the FCC lattice [3] reaches the same number through a fault-tolerant verification cost $E_s \times C_s = 36 \times 51 = 1836$.

The FCC unit cell contains two distinct interstitial void types: 8 tetrahedral voids (each bounded by 4 FCC vertices) and 4 octahedral voids (each bounded by 6 FCC vertices), with all bounding edges at the nearest-neighbor distance L (Section 2). The framework that traps a defect in the tetrahedral void simultaneously admits an analogous defect in the octahedral void — the same $K=4$ to $K=12$ phase transition, the same kinematic operators, the same geometric mechanism, applied to the second interstitial site that the FCC lattice provides. The natural question is what physics this companion defect predicts.

This paper develops the case that the octahedral-void defect is a viable candidate for dark matter. The case is built in two pieces. First, four qualitative properties of the defect follow rigorously from the structural symmetry of its bonding graph $K_{2,2,2}$ and bounding polyhedron (the regular octahedron with O_h symmetry): absence of first-order electromagnetic coupling, absence of $SU(3)$ color in the form generated for baryons, self-conjugate (Majorana-type) character, and suppressed first-order radiative cooling. These match the standard requirements for cold dark matter without invoking any free parameters. Second, the structural-counting framework of Ref. [2] that yields the proton’s verification cost $C_p = 1836$ extends to the octahedral defect by inclusion-exclusion on the $K_{2,2,2}$ bonding graph, terminating exactly at third order because the octahedron’s six vertices forbid any 4-matching: $C_{\text{DM}} = 25 \times 144 - 30 \times 10 + 8 \times 8 = 3364$. The framework therefore predicts the dark matter mass directly:

$$m_{\text{DM}} = \frac{C_{\text{DM}}}{C_p} \times m_p = \frac{3364}{1836} \times 938.272 \text{ MeV} = 1.719 \text{ GeV}, \quad (3)$$

using only the proton mass [8] as a calibration input. The recent detection of a ~ 1.5 – 1.6 GeV gamma-ray line in three active galactic nuclei [4] agrees with this prediction within 1 – 2.1σ of each individual source measurement, with the joint fit at ~ 1.55 GeV approximately 11% below the predicted rest-frame mass.

What this paper claims and does not claim. We present a forward derivation of the dark matter mass via inclusion-exclusion on the $K_{2,2,2}$ bonding graph (Section 4). The prediction $m_{\text{DM}} = 1.719$ GeV uses no cosmological observation as input and depends only on the bonded structure of the octahedral defect, the structural-counting framework of Ref. [2], and the proton mass.

We do not derive the cosmological abundance ratio $\Omega_{\text{DM}}/\Omega_b$ in this paper. The volume-trapping mechanism that fixes the relative formation rates of octahedral and tetrahedral defects, combined with the early-universe history of matter–antimatter annihilation in the tetrahedral sector and $\chi\chi \rightarrow \gamma\gamma$ pair-annihilation in the octahedral sector, relates the two abundances. A first-principles derivation requires quantitative treatment of both annihilation channels in the dense post-formation epoch, which is not attempted here. The forward mass prediction is independent of this calculation.

What we claim is:

1. The framework forces consideration of the octahedral-void defect as a second matter class. Its existence is not chosen; the crystallography of FCC requires both interstitial sites to be populated under any defect-generating mechanism that produces baryons from tetrahedral voids.
2. Four qualitative properties follow strictly from the defect’s geometric symmetry and match what dark matter must do.
3. The framework predicts $m_{\text{DM}} = 1.719$ GeV from the closed inclusion-exclusion expansion on the $K_{2,2,2}$ bonding graph, with no cosmological input. This is a forward prediction

derived from the same structural-counting machinery that yields $C_p = 1836$ for the proton in Ref. [2].

4. This mass prediction agrees with the recently reported 1.5–1.6 GeV gamma-ray line within $1\text{--}2.1\sigma$ of individual source measurements; the $\sim 11\%$ residual offset from the joint fit is consistent with gravitational redshift of inner-spike annihilation in the host AGN environment (Section 5).

Organization. Section 2 establishes the crystallographic geometry of the two interstitial void types in the FCC unit cell. Section 3 introduces the octahedral-void defect, defines its bonding graph $K_{2,2,2}$, and derives the four qualitative properties from the geometric symmetry. Section 4 presents the forward derivation of $m_{\text{DM}} = 1.719$ GeV via inclusion-exclusion on $K_{2,2,2}$, with explicit enumeration of all combinatorial inputs. Section 5 compares the prediction with current observational constraints, including the Fermi-LAT 1.5 GeV line, and contrasts the framework with existing dark matter models. Section 6 discusses falsifiability and open calculations. Section 7 concludes.

Interactive 3D Visualization. A WebGL visualization of the octahedral defect geometry — the $K_{2,2,2}$ bonding graph among the 6 bounding vertices, the 3 antipodal non-bonded pairs at second-nearest-neighbor distance $\sqrt{2}L$, and the cuboctahedral coordination cluster around any FCC vertex — accompanies this paper:

https://raghu91302.github.io/ssmtheory/oct_void_3D.html

The visualization makes immediately evident that the octahedral defect cannot be enclosed by any single 13-node coordination cluster: of the six bounding vertices, five lie within the cluster of any chosen anchor, but the sixth (the anchor’s antipode) lies outside.

2 Geometry of the Two Interstitial Voids

The FCC unit cell with cubic lattice constant a contains atoms at the cube corners and face centers, giving a Bravais lattice with primitive cell volume $a^3/4$ and nearest-neighbor distance $L = a/\sqrt{2}$. The unit cell decomposes into two distinct interstitial void types [9, 10] (Figure 1).

Tetrahedral voids. Located at $(a/4)(\pm 1, \pm 1, \pm 1)$ with all eight sign combinations, giving 8 voids per cell. Each is bounded by 4 FCC vertices forming a regular tetrahedron with edge length L . The centroid-to-vertex distance is $L\sqrt{3}/8 \approx 0.612L$.

Octahedral voids. Located at the body center $(a/2, a/2, a/2)$ and at the 12 edge midpoints. Each edge midpoint is shared between 4 unit cells, contributing $12/4 = 3$ to the cell count, plus 1 body-center, giving 4 voids per cell. Each is bounded by 6 FCC vertices forming a regular octahedron with edge length L . The centroid-to-vertex distance is $L/\sqrt{2} \approx 0.707L$.

3 The Octahedral-Void Defect and Its Structural Properties

3.1 The defect: trapped K=6 remnant

During the $K=4 \rightarrow K=12$ phase transition that crystallizes the SSM vacuum [2], the same kinematics that traps a $K=4$ remnant in a tetrahedral void can trap a remnant in an octahedral void. The trapped node sits at the centroid of the octahedral void and bonds to the six bounding vertices, creating a local $K=6$ pocket inside the $K=12$ bulk. This parallels the proton’s $K=4$ trapped remnant in the tetrahedral case.

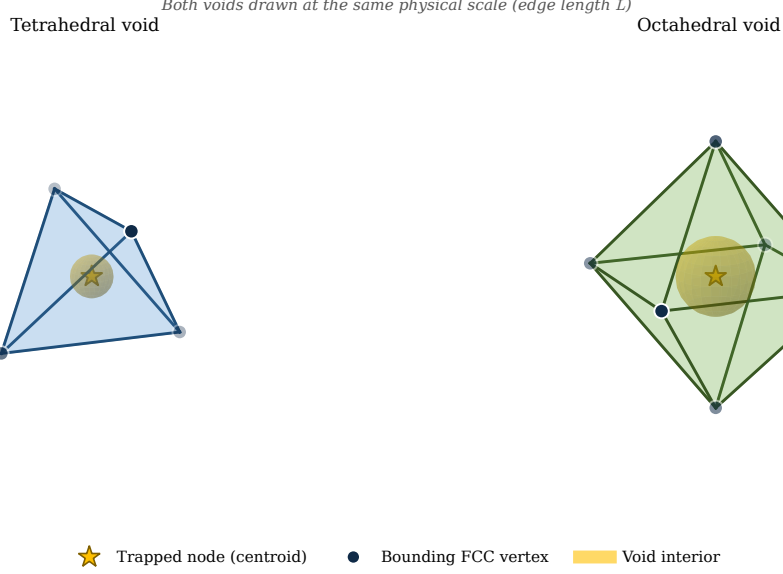


Figure 1: The two interstitial void types of the FCC unit cell. **Left:** A tetrahedral void, bounded by 4 FCC vertices (dark blue) forming a regular tetrahedron with edge length L . The trapped $K=4$ node (yellow star) sits at the centroid; the yellow translucent sphere illustrates the void interior. **Right:** An octahedral void, bounded by 6 FCC vertices forming a regular octahedron with the same edge length L . The trapped $K=6$ node sits at the centroid; the yellow translucent sphere illustrates the void interior. The octahedral void is geometrically larger than the tetrahedral void, reflecting the longer centroid-to-vertex distance ($L/\sqrt{2}$ versus $L\sqrt{3}/8$).

Why $K=4$ and $K=6$, and not intermediate values? The selection of the tetrahedral ($K=4$) and octahedral ($K=6$) configurations as the only viable trapped-remnant classes follows from two independent constraints that coincide in the FCC lattice.

Crystallographic constraint. The only positions in the FCC vacuum that admit multiple FCC nodes at the unit nearest-neighbor distance L are the tetrahedral interstitial site (4 equidistant nearest neighbors) and the octahedral interstitial site (6 equidistant nearest neighbors). Generic positions in FCC have a unique closest FCC node and cannot host a trapped configuration with multiple equal-length bonds. There is no position in FCC with exactly 5 (or 7, 8, etc.) equidistant nearest neighbors. The interstitial sites of the perfect FCC lattice therefore offer only $K=4$ and $K=6$ as candidate trapping geometries.

Strain-balance constraint. Even hypothetically considering 5 of the 6 vertices of an octahedral void as the bounding set of a square-pyramidal defect, the trapped node would not sit at the geometric centroid in mechanical equilibrium. The inward unit-vector sum $\sum_i \hat{n}_i$ from each bounding vertex toward the centroid vanishes identically for the regular tetrahedron and the regular octahedron, by their inversion or rotational symmetries. For any subset that breaks this symmetry — including a square pyramid (5 vertices) — $\sum_i \hat{n}_i \neq 0$, and the trapped node’s strain-energy minimum is displaced along the symmetry-breaking axis (toward or away from the removed vertex). Such a defect would be intrinsically asymmetric, carrying a built-in structural dipole, and is not produced by the isotropic kinematics of the $K=4 \rightarrow K=12$ phase transition.

The two constraints reinforce each other: the symmetric strain-balanced configurations are exactly the symmetric crystallographic interstitial voids. The SSM framework therefore naturally selects $K=4$ and $K=6$ as the only stable trapped-remnant classes, with no fitting required and no intermediate- K particle classes admitted.

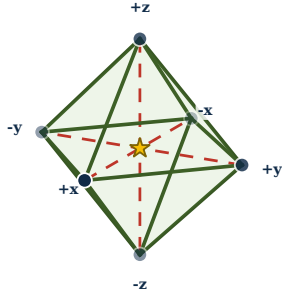
3.2 Bonding subgraph: $K_{2,2,2}$

The 6 bounding vertices of the octahedron sit at three antipodal pairs: each pair at distance $L\sqrt{2}$, and each non-antipodal pair at distance L (the nearest-neighbor distance). Bonds at the SSM unit-bond-length scale L connect each vertex to its 4 non-antipodal partners, generating the complete tripartite graph

$$K_{2,2,2} : \quad 6 \text{ vertices,} \quad 12 \text{ edges,} \quad \text{degree 4 per vertex,} \quad 3 \text{ antipodal non-bonded pairs.} \quad (4)$$

This is the bonded subgraph of the octahedral-void defect (Figure 2), in contrast to the proton case where the bonded subgraph is K_4 (4 vertices, 6 edges, degree 3).

Bond graph $K_{2,2,2}$: 12 unit-length edges (green),
3 antipodal non-bonds at $L\sqrt{2}$ (red dashed)



Face structure: schematic of all 8 triangular faces

All 8 faces are triangular (C_3 , non-bipartite)



0 square faces $\Rightarrow$ no C_4 bipartite EM channel

Figure 2: Bonded substructure of the octahedral defect. **Left:** The 6 bounding vertices of the octahedron with the trapped $K=6$ node (yellow star) at the centroid. The 12 unit-length bonds form the complete tripartite graph $K_{2,2,2}$ (solid green); the 3 antipodal pairs at distance $L\sqrt{2}$ are non-bonded under the SSM unit-bond rule (red dashed). **Right:** The bounding polyhedron has 8 triangular faces and 0 square faces. The absence of square (C_4) faces is the structural origin of the suppressed first-order electromagnetic coupling discussed in Section 3.3.

3.3 Suppressed first-order electromagnetic coupling

The argument for the photon coupling channel of baryonic defects given in Ref. [2] proceeds in two steps. First, the defect's coupling to the bulk lattice is mediated by oscillation modes hosted on the bounding faces of its internal bonded structure: triangular faces (cycle C_3) cannot carry the bipartite alternation that supports a dipolar mode, while square faces (cycle C_4) can. Second, the bulk cuboctahedral coordination shell of each FCC vertex contains 8 triangular faces and 6 square faces, providing the propagation channel through which the defect's bipartite oscillation modes couple to bulk photon modes.

For the tetrahedral-void defect, the bounding polyhedron is a regular tetrahedron with 4 triangular faces and zero square faces. The photon channel of Ref. [2] is not strictly forbidden because the dipolar oscillation can be carried on the square faces of the surrounding bulk cuboctahedral shells, which the defect's external bonds traverse.

For the octahedral-void defect, the bounding polyhedron is a regular octahedron with 8 triangular faces and *zero* square faces, and the bonded subgraph $K_{2,2,2}$ contains 4-cycles only between antipodal pairs that are explicitly non-bonded. The defect itself has no internal C_4 structure on its bonded subgraph and is bounded by 8 triangular faces. The defect-to-bulk

coupling that would mediate first-order photon emission has no preferred bipartite mode at the defect side: the first-order EM coupling channel is suppressed at the defect-internal level.

Falsifiable consequence. An octahedral-void defect interacts with the electromagnetic field only through higher-order channels (multi-photon processes, mediation through virtual baryonic intermediaries, or gravitational coupling to bulk photon modes). Direct first-order single-photon coupling is suppressed by the absence of bipartite face structure on the defect’s bounding polyhedron. This matches the standard dark-matter requirement of EM neutrality at the level of leading-order cross-sections.

Direct-detection bounds on dark-matter–photon couplings (currently constraining the dark photon mixing parameter $\epsilon \lesssim 10^{-3}$ at ~ 1 GeV from beam-dump experiments and supernova cooling [14, 15]) are consistent with the picture predicted here. A complete proof of EM-neutrality requires deriving the photon coupling vertex from the SSM strain-field dynamics; the argument given here is the structural-symmetry analog of the Ref. [2] argument applied to the octahedral case.

3.4 Absence of SU(3) color in the form generated for baryons

In the tetrahedral case, three color charges arise from the three skew-edge pairs of the bounding tetrahedron K_4 [2]. The combinatorial identity $C(4, 2)/2 = 3$ generates the three-color SU(3) representation on baryon-like defects.

For the octahedral case, the skew-edge pair count of $K_{2,2,2}$ is

$$c_{\text{skew}}^{(O)} = \binom{12}{2} - 6\binom{4}{2} = 66 - 36 = 30. \quad (5)$$

This count is verified by direct enumeration: each of the 12 edges of $K_{2,2,2}$ has exactly 5 skew partners (the edges connecting the 4 remaining vertices, minus the one antipodal non-bond), giving $12 \times 5 / 2 = 30$ skew pairs. It is also obtained by counting the 2-matchings of the octahedron graph.

The skew-pair count 30 does not factor as a small representation analogous to the three-color case. The natural color structure on an octahedral defect is therefore not SU(3) of the standard QCD form. The defect does not couple through the three-color confining channels of the cuboctahedral shell.

Falsifiable consequence. An octahedral-void defect does not participate in standard strong-force interactions. It cannot bind to baryons through gluon exchange, cannot form bound states with quarks, and cannot hadronize. This matches the standard dark-matter requirement of strong-force neutrality.

3.5 Self-conjugate (Majorana-type) character

Reference [2] identifies anti-baryons with spatial inversion of the trapped tetrahedral-void defect: the FCC unit cell contains two distinct tetrahedral-void orientations (centered at $(a/4, a/4, a/4)$ and $(3a/4, 3a/4, 3a/4)$), related by inversion $\hat{r} \rightarrow -\hat{r}$ that exchanges the sign of the anchor projection. The two orientations support matter and anti-matter respectively; the cosmological matter–antimatter asymmetry corresponds to the local cosmological excess of one orientation over the other.

The octahedral void has no analogous orientation degeneracy. The regular octahedron is invariant under inversion through its centroid: the symmetry group O_h contains the inversion

operator $-I$, which maps the octahedron to itself rather than to a distinct partner. Each octahedral void (body-center or edge-midpoint) is therefore its own image under inversion. The framework predicts that the octahedral-void defect is its own anti-particle: a stable, self-conjugate Majorana-type dark-matter particle.

Falsifiable consequences.

- No matter–antimatter asymmetry is required (or possible) in the dark sector. The cosmological abundance of the dark species is set by formation rates and the early-universe history of pair-annihilation, not by an asymmetry between particle and antiparticle.
- Pair-annihilation of two octahedral defects is in principle allowed ($\chi\chi$, not $\chi\bar{\chi}$), with rate set by the available coupling channels. With suppressed first-order EM, annihilation rates are loop-suppressed. This is structurally consistent with the loop-suppressed $\langle\sigma v\rangle_{\gamma\gamma} \approx 2 \times 10^{-28} \text{ cm}^3/\text{s}$ required by the Kang et al. 2026 AGN line [4].
- Searches for a Dirac-type dark matter particle with distinct anti-particle (e.g., dark-asymmetric models) would not find a match in this framework.

3.6 Suppressed first-order radiative cooling and clustering asymmetry

A central observational distinction between baryonic and dark matter is the difference in their large-scale clustering behavior. Both species clump under gravity — galaxies, clusters, and the cosmic web are dark-matter-dominated gravitational structures, with baryons collected in their potential wells. The asymmetry is in how each species behaves *after* gravitational infall: baryons collapse to high densities through radiative cooling (atomic line emission, bremsstrahlung, free-free emission), forming stars, planets, and dense compact objects, while dark matter remains diffuse on scales below the cluster halo, with no equivalent of stellar or planetary collapse.

The framework predicts this asymmetry from the same structural feature already invoked for EM-suppression. Tetrahedral-void defects (baryons) couple to photons through the bulk cuboctahedral coordination shell, providing a first-order radiative channel through which kinetic energy can be shed. Octahedral-void defects, by contrast, lack the defect-internal bipartite structure required to source the dipolar oscillation modes that mediate first-order photon emission. The same suppression that yields EM-neutrality also suppresses radiative cooling.

A non-radiatively-cooling gravitating species is collisionless on astrophysical timescales: it cannot shed kinetic energy efficiently, cannot collapse below its initial velocity-dispersion scale, and forms diffuse halos rather than dense compact objects. This matches the observed phenomenology of dark matter, including the diffuse extension of galactic and cluster halos beyond the visible baryonic component, and the absence of dark stellar or planetary analogues at any scale. The recent identification of “almost dark” galaxies dominated by dark matter halos with negligible stellar content provides further phenomenological support; we discuss this class of observation in Section 5.4. The prediction is therefore that octahedral defects are collisionless on cosmological timescales relative to baryonic radiative cooling rates, consistent with current observational constraints on dark-matter self-interaction [11, 12, 13].

4 Forward Derivation of C_{DM} via Inclusion-Exclusion

The structural-counting derivation of $m_p/m_e = 1836$ for the proton presented in Ref. [2] (Eq. (2)) extends to the octahedral defect by inclusion-exclusion on the $K_{2,2,2}$ bonding graph. We present this extension here and obtain the framework’s forward prediction $C_{\text{DM}} = 3364$, $m_{\text{DM}} = 1.719 \text{ GeV}$, with no cosmological input.

4.1 The structural-counting formula as inclusion-exclusion

The expression of Ref. [2], $C_p = (K+1)K^2 - c_{\text{skew}}K = 13 \times 144 - 3 \times 12 = 1836$, admits a natural interpretation in the language of the Principle of Inclusion-Exclusion (PIE). Each structural node of the defect carries a disruption halo of $K^2 = 144$ vacuum bond states (the second-shell footprint at the bulk coordination $K = 12$). Summed over $N_T = 13$ structural nodes, this gives a base count of $13 \times 144 = 1872$ bond-state disruptions. Pairs of structural nodes with mutually disjoint flux channels — the $c_{\text{skew}} = 3$ skew-edge pairs of K_4 — have disruption halos that overlap, double-counting bond states. The size of each pairwise overlap, computed directly as the bulk first-shell intersection $|\mathcal{N}(e_1) \cap \mathcal{N}(e_2)|$ on the FCC lattice, is $K_{\text{pairwise}} = 12$ for any K_4 skew pair. Subtracting the double-counted bond states gives $1872 - 36 = 1836$, the proton's verified verification cost. The series terminates: K_4 has only 4 vertices, so it admits no triple of mutually disjoint edges (a 3-matching requires 6 vertices), and the third-order PIE term is identically zero.

4.2 Extension to $K_{2,2,2}$ with truncation

The same structure applies to the octahedral defect, with three differences fixed by the bonded graph $K_{2,2,2}$:

1. **Structural node count.** Each of the 6 bounding vertices contributes 4 bonds within $K_{2,2,2}$, giving $6 \times 4 = 24$ boundary structural nodes; with the trapped center, $N_O = 25$.
2. **Skew-edge pair count.** Each of the 12 edges of $K_{2,2,2}$ has exactly 5 mutually disjoint partners (12 total edges minus 1 self minus 6 edges sharing a vertex), giving $c_{\text{skew}}^{(O)} = (12 \times 5)/2 = 30$ skew pairs. The same count is obtained by direct enumeration of 2-matchings of the octahedron graph and by the closed form $\binom{12}{2} - 6\binom{4}{2} = 66 - 36 = 30$.
3. **Pairwise overlap on $K_{2,2,2}$.** Computing the bulk first-shell intersection directly on the FCC lattice yields $|\mathcal{N}(e_1) \cap \mathcal{N}(e_2)| = 10$ uniformly across all 30 skew pairs of $K_{2,2,2}$, in contrast to K_4 's value of 12. The pairwise overlap is a graph-specific geometric quantity, not a fixed bulk constant.

Crucially, $K_{2,2,2}$ admits the next term in the PIE expansion that K_4 does not. With 6 vertices, the octahedron graph hosts triples of mutually disjoint edges (3-matchings, equivalently perfect matchings of the octahedron), of which there are exactly $c_{\text{triple}}^{(O)} = 8$ (verified by direct enumeration; see Section 4.3). By PIE, the triple-overlap contribution must be added back to compensate for the pairwise subtraction's overcounting in regions where three flux channels mutually intersect.

The triple overlap on $K_{2,2,2}$, computed directly as the bulk first-shell triple intersection $|\mathcal{N}(e_1) \cap \mathcal{N}(e_2) \cap \mathcal{N}(e_3)|$, evaluates to $K_{\text{triple}}^{(O)} = 8$ uniformly across all 8 perfect matchings of the octahedron.

The series terminates exactly at third order. A 4-matching requires 8 mutually distinct vertices; the octahedron graph has only 6. Therefore $c_{\text{quad}}^{(O)} = 0$ and all higher orders vanish identically. The PIE expansion for the octahedral defect is finite and closed:

$$\begin{aligned} C_{\text{DM}} &= N_O \cdot K_{\text{bulk}}^2 - c_{\text{skew}}^{(O)} \cdot K_{\text{pairwise}}^{(O)} + c_{\text{triple}}^{(O)} \cdot K_{\text{triple}}^{(O)} \\ &= 25 \times 144 - 30 \times 10 + 8 \times 8 \\ &= 3600 - 300 + 64 = \mathbf{3364}. \end{aligned} \tag{6}$$

The forward prediction from this derivation is $m_{\text{DM}} = (C_{\text{DM}}/C_p) \times m_p = (3364/1836) \times 938.27 \text{ MeV} = 1.719 \text{ GeV}$.

4.3 Explicit enumeration of pairwise and triple overlaps

The values $K_{\text{pairwise}}^{(O)} = 10$ and $K_{\text{triple}}^{(O)} = 8$ appearing in Eq. (6) are not free parameters: they are determined by direct enumeration on the FCC lattice with no fitting. We make the enumeration explicit here so a reader can verify the computation without recourse to external code.

First shell of an FCC vertex. Each FCC node v has exactly 12 nearest neighbors (its cuboctahedral coordination shell), at the 12 displacements

$$\mathcal{N}_1(v) - v \in \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\} \quad (7)$$

in lattice units (parity-even FCC convention with NN distance $L = \sqrt{2}$). For a $K_{2,2,2}$ edge $e = \{v_a, v_b\}$ between two such vertices, we define the first-shell neighborhood as the union

$$\mathcal{N}(e) = \mathcal{N}_1(v_a) \cup \mathcal{N}_1(v_b), \quad (8)$$

which contains $|\mathcal{N}(e)| = 12 + 12 - 4 = 20$ nodes for any edge in $K_{2,2,2}$ (the 4 subtracted are the four common nearest neighbors of v_a and v_b).

Pairwise overlap. For each of the 30 skew-edge pairs (e_i, e_j) in $K_{2,2,2}$, the pairwise intersection is

$$|\mathcal{N}(e_i) \cap \mathcal{N}(e_j)| = 10, \quad (9)$$

uniformly across all 30 pairs. This is verified by direct enumeration on the FCC lattice; the value 10 emerges from the geometry of the octahedral void (with the void center at $(1, 0, 0)$ and bounding vertices at $\{(2, 0, 0), (0, 0, 0), (1, \pm 1, 0), (1, 0, \pm 1)\}$) and is invariant under permutation of the three coordinate axes. For comparison, on the K_4 tetrahedral void the analogous quantity yields $|\mathcal{N}(e_i) \cap \mathcal{N}(e_j)| = 12$ uniformly across the 3 skew pairs — the value used implicitly in the proton derivation of Ref. [2]. The two values (10 vs. 12) reflect the different bonding-graph geometry; neither is a free parameter.

Triple overlap. For each of the 8 perfect matchings (e_i, e_j, e_k) of $K_{2,2,2}$, the triple intersection is

$$|\mathcal{N}(e_i) \cap \mathcal{N}(e_j) \cap \mathcal{N}(e_k)| = 8, \quad (10)$$

uniformly across all 8 matchings. The triple intersection set decomposes structurally as

$$\underbrace{6}_{\text{bounding vertices}} + \underbrace{2}_{\text{matching-specific bulk nodes}} = 8. \quad (11)$$

The 6 bounding vertices of the octahedral void are common to the intersection of *every* matching — they are the lattice nodes adjacent to all 12 octahedron edges by construction. The remaining 2 nodes vary between matchings and lie at the FCC nodes most strongly correlated with the specific orientation of the chosen 3-matching.

The $8 = 8$ coincidence. Equation (6) contains two distinct quantities that both equal 8:

- $c_{\text{triple}}^{(O)} = 8$, the number of perfect matchings of the octahedron graph $K_{2,2,2}$. This is a graph-theoretic invariant, independent of the embedding lattice.
- $K_{\text{triple}}^{(O)} = 8$, the cardinality of the bulk first-shell triple intersection, computed on the FCC lattice for any one such matching. This is a geometric quantity, dependent on both the bonding graph and the embedding lattice.

These quantities are independent. The graph-theoretic 8 (matching count) arises from the symmetry $K_{n,n,n}$ at $n = 2$. The geometric 8 (overlap size) arises from the FCC neighborhood structure and decomposes as $6 + 2$ as above. The numerical equality is a coincidence of the specific combination of $K_{2,2,2}$ bonding graph and FCC lattice; either factor would change under deformation of the lattice neighborhood, while the matching count would not. We emphasize that Eq. (6) multiplies these two as independent inputs and does not double-count a single combinatorial quantity.

Robustness check. A useful sanity check is that the proton derivation of Ref. [2] arises as the $K = K_{\text{bulk}}$, $N \rightarrow N_T = 13$, $c_{\text{skew}} \rightarrow 3$ specialization of the same PIE structure with $K_{\text{pairwise}} = K_{\text{bulk}} = 12$ on K_4 . The series terminates at second order there because K_4 admits no 3-matching. The transition from proton ($K_{\text{pairwise}} = 12$) to octahedral defect ($K_{\text{pairwise}}^{(O)} = 10$) is the only place where a graph-specific geometric quantity replaces a bulk constant in the verification-cost formula; this replacement is forced by the geometry of $K_{2,2,2}$ and is the principal new combinatorial input of the present paper.

Verification artifact. The full enumeration tables for all 30 pairwise intersections and all 8 triple intersections (with explicit $6 + 2$ decomposition) are provided in the Appendix, and a short self-contained Python reference implementation that reproduces every integer in Eq. (6) from the FCC nearest-neighbor vectors and the six bounding vertex coordinates alone is available at https://github.com/raghu91302/ssmtheory/blob/main/verify_C_DM.py. The script uses only the Python standard library and runs in under one second; a reader who wishes to verify $K_{\text{pairwise}}^{(O)} = 10$ and $K_{\text{triple}}^{(O)} = 8$ independently can do so either by inspection of the tables or by execution of the script.

4.4 Embedding Uniformity Lemma and Body-Diagonal Structure

The values $K_{\text{pairwise}}^{(O)} = 10$ and $K_{\text{triple}}^{(O)} = 8$ were established in Section 4.3 by direct enumeration. A reader is entitled to ask whether the observed *uniformity* across the 30 skew pairs and the 8 perfect matchings is itself a structural fact or a coincidence of independent computations. The following lemma settles this question and identifies a clean combinatorial labeling of the 8 matchings by the 4 body diagonals of the cube circumscribing the octahedral void.

Lemma (Embedding Uniformity). *Embed the octahedral void at the origin of the FCC lattice as in Section 4.3, and let O_h denote its point stabilizer (the symmetry group of the regular octahedron, of order 48). Then:*

- (i) *The 8 perfect matchings of $K_{2,2,2}$ form a single O_h -orbit. The triple first-shell intersection size $|\mathcal{N}(e_1) \cap \mathcal{N}(e_2) \cap \mathcal{N}(e_3)|$ is therefore constant across all matchings, and equals 8 by direct evaluation on any representative matching. Hence $K_{\text{triple}}^{(O)} = 8$ is O_h -forced.*
- (ii) *The 30 skew pairs of $K_{2,2,2}$ split into two O_h -orbits, of sizes 24 and 6. The pairwise first-shell intersection size is constant within each orbit by symmetry; direct evaluation on a representative of each orbit gives the value 10 in both cases. Hence $K_{\text{pairwise}}^{(O)} = 10$ uniformly.*

Status of (i) versus (ii). The triple statement (i) is fully O_h -forced: a single orbit means the symmetry alone determines uniformity. The pairwise statement (ii) is stronger than O_h -symmetry alone can force. With two orbits, O_h -invariance guarantees uniformity *within* each orbit but allows the two orbits to give different values; direct enumeration is required to confirm that they happen to give the same value 10. We record this as an empirical fact about the

particular embedding of $K_{2,2,2}$ in the FCC lattice rather than as a consequence of pure symmetry. The orbit decomposition and both intersection-size evaluations are verified by the standalone script https://github.com/raghu91302/ssmtheory/blob/main/verify_uniformity.py.

Body-diagonal labeling of the matchings. The 8 perfect matchings admit a natural geometric labeling by the 4 body diagonals of the cube circumscribing the octahedral void — the cube whose 8 corners sit at displacement $(\pm 1, \pm 1, \pm 1)$ from the void center, and whose 6 face centers are the bounding vertices A, B, C, D, E, F of the octahedron. For each matching M :

- (a) The 2 “matching-specific” bulk nodes appearing in the $6 + 2$ decomposition of the triple intersection (Table 4) are exactly the two antipodal corners of one body diagonal of this cube. Each matching is therefore labeled by a unique body diagonal.
- (b) Each body diagonal labels exactly two matchings; the 8 matchings group into 4 pairs in bijection with the 4 body diagonals.
- (c) Within each such pair, the two matchings are related by inversion through the void center, $\mathbf{r} \mapsto -\mathbf{r}$ (in coordinates relative to the void center). This is the identical centrosymmetry $-I \in O_h$ that underlies the self-conjugate (Majorana-type) character of the defect established in Section 3.5.
- (d) The three edge midpoints of any matching M lie in the plane through the void center perpendicular to M ’s labeling body diagonal.

These four properties are verified by the same standalone script.

Implication for the “8 = 8” coincidence. The factorization $c_{\text{triple}}^{(O)} \cdot K_{\text{triple}}^{(O)} = 8 \times 8 = 64$ in Eq. (6) multiplies two structurally independent quantities (Section 4.3, “8 = 8 coincidence”): the graph-theoretic 8 (number of perfect matchings of $K_{2,2,2}$, fixed by the symmetry $K_{n,n,n}$ at $n = 2$) and the geometric 8 (size of the bulk first-shell triple intersection, fixed by the FCC embedding). The body-diagonal labeling gives each of these 8’s an independent structural origin — the matchings are in bijection with the 4 cube diagonals via the inversion pairing, and the triple intersection decomposes as $6 + 2$ where the 2 are precisely the diagonal’s endpoints. The numerical equality of the two 8’s in this specific embedding remains a non-trivial fact about $K_{2,2,2}$ in FCC, but it is no longer an unstructured coincidence: each side has its own geometric reading.

4.5 QEC dual: an open question

The structural counting result $C_p = 1836$ of Ref. [2] has an independent realization as a fault-tolerant verification cost $E_s \times C_s = 36 \times 51 = 1836$ in a CSS code on the FCC lattice [3]. That construction is built on the 13-node cuboctahedral coordination cluster around a single FCC node, and the proton’s tetrahedral defect fits inside this single cluster: its 4 bounding vertices are mutually at the nearest-neighbor distance L .

The octahedral defect’s 6 bounding vertices include 3 antipodal pairs at second-nearest-neighbor distance $\sqrt{2}L$, which prevents the defect from being captured by any single coordination cluster and requires a multi-cluster QEC footprint. Constructing the QEC dual of Eq. (6) is left to future work. We note that the truncation of the PIE series at third order, by the 6-vertex constraint of $K_{2,2,2}$, suggests that the QEC dual should likewise be a closed expression rather than an asymptotic series.

4.6 Status of the forward derivation

The qualitative predictions of Section 3 stand independently of the verification-cost calculation: the absence of first-order EM coupling, absence of SU(3) color, self-Majorana character, and suppressed radiative cooling all follow from the structural symmetry of the $K_{2,2,2}$ bonding graph and the bounding octahedron. The quantitative mass prediction in this paper is $m_{\text{DM}} = 1.719$ GeV, derived from the closed PIE expansion (Eq. (6)) and the proton mass alone.

5 Comparison with Observations

5.1 The Fermi-LAT 1.5–1.6 GeV gamma-ray line

A recent Fermi-LAT analysis [4] reports a gamma-ray spectral line at approximately 1.5–1.6 GeV in three active galactic nuclei (AGN), with combined significance $> 5\sigma$ (joint test statistic $TS = 57.77$). The individual source measurements and the SSM forward mass prediction $m_{\text{DM}} = 1.719$ GeV (Eq. (6)) are compared in Table 1.

Source (4FGL designation)	E_{line} (GeV)	Deviation from 1.719 GeV	$ \Delta /\sigma_E$
J0749.6+1324 (blazar)	1.62 ± 0.07	+0.099 GeV	1.4
J0250.2–8224 (blazar)	1.55 ± 0.10	+0.169 GeV	1.7
J2329.7–2118 (radio galaxy)	1.53 ± 0.09	+0.189 GeV	2.1
J0357.0–4955 (marginal candidate)	1.50 ± 0.21	+0.219 GeV	1.0
Joint fit	~ 1.55	+0.169 GeV	$\sim 11\%$

Table 1: Fermi-LAT line position measurements [4] compared with the SSM forward mass prediction $m_{\text{DM}} = 1.719$ GeV (assuming $\chi\chi \rightarrow \gamma\gamma$ annihilation, in which case $E_{\text{line}} = m_{\text{DM}}$). The first three sources are reported as primary detections by [4] with line-detection significances 2.8–4.1 σ ; J0357.0–4955 is a marginal candidate at 1.6 σ detection significance and is included here for completeness. The fourth column shows the deviation of the measured line position from the SSM prediction, expressed in units of the line measurement uncertainty σ_E (*not* the detection significance). The forward prediction agrees with each individual source measurement within 1–2.1 such units. The joint fit at ~ 1.55 GeV is approximately 11% below the prediction, slightly outside Fermi-LAT’s $\sim 10\%$ instrumental energy resolution at 1.5 GeV; the residual offset is discussed in Section 5.

The match should be interpreted carefully. The Kang et al. 2026 line, while detected at $> 5\sigma$ in joint analysis, remains at ~ 2.8 –4.1 σ in individual sources, and the Galactic Center analysis “did not converge” in the discoverers’ work [4]. Confirmation requires additional observations and analysis. We note three features that connect the Fermi-LAT signal to the SSM prediction beyond the mass agreement.

Cross-section structure. The Kang et al. 2026 fit requires $\langle\sigma v\rangle_{\gamma\gamma} \approx 2 \times 10^{-28}$ cm³/s, approximately 150 $\times$ smaller than the canonical thermal-relic WIMP cross-section. The discoverers attribute this to one-loop EM suppression of the $\chi\chi \rightarrow \gamma\gamma$ channel. The SSM prediction of suppressed first-order EM coupling (Section 3.3) is structurally consistent with this loop-suppression: a particle whose tree-level photon coupling is absent or strongly suppressed will produce loop-suppressed annihilation rates without requiring the EM-suppression to be put in by hand.

Self-conjugate annihilation. The Kang et al. 2026 analysis treats the line as $\chi\chi$ annihilation (not $\chi\bar{\chi}$), consistent with the SSM prediction that the octahedral defect is its own antiparticle

(Section 3.5). Models with distinct particle/antiparticle would require additional channels or assumptions to fit the same data.

Source environment and gravitational redshift. The Fermi-LAT line is observed in active galactic nuclei, which host supermassive black holes (SMBHs) and are predicted to develop dark-matter density spikes through adiabatic contraction [26]. Because annihilation flux scales as ρ^2 , emission is dominated by material at small radii in the spike. Photons emerging from this inner region experience gravitational redshift by a factor $\sqrt{1 - R_s/R}$, where R_s is the SMBH’s Schwarzschild radius. For emission at $R \approx 5 R_s$, the redshift factor is ≈ 0.90 , mapping a rest-frame line at 1.719 GeV to ~ 1.55 GeV observed — consistent with the joint fit of [4]. The individual sources tabulated (three primary detections plus the marginal candidate), with observed energies spanning 1.50–1.62 GeV, would correspond to effective emission radii in the range $R \approx 4\text{--}9 R_s$, plausibly reflecting source-to-source variation in spike profile, SMBH spin, or line-of-sight geometry. We emphasize this is a candidate astrophysical interpretation, not a derivation: the same redshift mechanism would map any sufficiently heavier rest-frame mass to the observed line for a suitably chosen emission radius. A more discriminating test would compare predicted line shapes (broadened by the spike’s radial extent and by the orbital velocity dispersion of the annihilating particles) with observed source spectra; we identify this as an additional observational direction for future work.

Scope of agreement. The mass agreement at the $1\text{--}2.1\sigma$ level per source, with a $\sim 11\%$ residual on the joint fit, does not by itself establish that the line is dark matter of the SSM type. Two cautions deserve explicit statement. First, the most direct historical precedent for a Fermi-LAT gamma-ray line claim at comparable per-source significance is the ~ 130 GeV feature identified by Bringmann et al. and Weniger in 2012 [5, 6], which generated extensive theoretical follow-up before being substantially attenuated by additional data and reprocessed instrument response functions and ultimately attributed in significant part to instrumental systematics [7]. The Kang et al. 2026 detection is at a different energy and in a different source class (extragalactic AGN rather than Galactic Center), so the instrumental story is not the same; but the base rate for line claims at $\sim 3\text{--}4\sigma$ per source surviving further data is poor, and confirmation by an independent analysis or instrument is required before the present mass agreement carries decisive weight. Second, the gravitational-redshift mechanism that closes the $\sim 11\%$ residual is itself a free degree of freedom: as noted above, an emission radius $R \approx 5 R_s$ produces the observed shift, but the same mechanism (which is strictly one-sided, observed $\leq$ rest-frame) would accommodate any rest-frame mass between the observed line and $\sim 1.2\times$ the observed line by an appropriate choice of R , covering the entire range plausibly populated by spike-dominated emission. The mass-prediction falsifiability is therefore $O(20\%)$ once the redshift is left unconstrained; a sharper test requires either an independent spike-profile measurement at one of the AGN or a line-shape comparison of the kind noted at the end of the preceding paragraph. What Table 1 does establish is that the framework’s forward prediction (derived from the structural-counting expansion on $K_{2,2,2}$, with no cosmological input) is consistent with the principal observational anchor available, and that confirmation or refutation of the Fermi-LAT signal in future work will directly bear on the framework.

5.2 Direct detection

A 1.7 GeV particle with suppressed first-order EM and no strong coupling has recoil energies in the few-keV range when scattering off nuclei via available higher-order channels. Standard WIMP detectors (xenon-based TPCs, germanium detectors) have thresholds in the few-keV range and are optimized for higher-mass candidates. Direct detection by current experiments at WIMP scales [21, 22] is not expected for this mass with this coupling structure. Future

light-dark-matter detectors (DAMIC, SENSEI, low-threshold cryogenic detectors) operating in the sub-keV range may achieve sensitivity if any non-gravitational coupling persists through higher-order channels.

5.3 Self-interaction cross-section

Two octahedral-void defects passing close to each other interact through their respective strain fields in the FCC bulk. A leading-order geometric estimate (cross-section $\sim \pi(2L)^2 = 4\pi L^2$, scaled by mass to give σ/m) yields $\sigma/m \sim 0.01\text{--}0.1 \text{ cm}^2/\text{g}$ for L in the QCD-scale range 0.5–2 fm. This places the framework in the “essentially collisionless” regime, consistent with the Bullet Cluster bound $\sigma/m \lesssim 0.47 \text{ cm}^2/\text{g}$ [13] and with the absence of strong dark-sector self-interaction signatures in cluster collisions.

The leading-order estimate sits below the parameter window $\sigma/m \approx 0.1\text{--}2 \text{ cm}^2/\text{g}$ [20, 19] where self-interacting dark matter (SIDM) models address small-scale-structure observations. The framework as constituted predicts cold dark matter that is largely collisionless on observational scales, with a small residual self-interaction below current detection sensitivity. A more precise calculation requires defining the strain field of the octahedral defect quantitatively, which is left to future work.

5.4 Diffuse dark-matter halos and almost-dark galaxies

The SSM prediction in Section 3.6 that octahedral-void dark matter cannot dissipate energy at first order, and therefore forms diffuse halos rather than collapsed compact structures, is observationally testable through the existence of dark-matter-dominated galaxies with extended, low-surface-brightness morphologies and minimal stellar content.

The recent identification of Candidate Dark Galaxy-2 (CDG-2) in the Perseus cluster [23] provides an observational example of this phenomenology. CDG-2 was first detected through globular-cluster overdensity rather than diffuse stellar emission — the standard search methodology that fails for galaxies whose stellar component is too faint to detect directly. Subsequent stacking of HST imaging and analysis of Euclid Early Release Observations revealed extremely faint but significant diffuse emission consistent across both datasets, validating CDG-2 as a genuine galaxy. The system contains four globular clusters within a ~ 1.2 kpc diameter and a total galaxy luminosity $L_{V,\text{gal}} \approx 6.2 \times 10^6 L_\odot$, with the GC light fraction estimated at $\sim 16.6\%$ (rising to $\sim 33\%$ if a canonical GC luminosity function is assumed). Applied to the GC-to-halo-mass scaling relations of [24] and [25], these data imply a dark-matter halo mass fraction of 99.94–99.99%, making CDG-2 plausibly the most dark-matter-dominated galaxy yet identified.

This is exactly the morphological signature predicted by a non-radiatively-cooling dark sector: the dark-matter halo is extended and diffuse, the baryonic component is minimal and confined to the residual stellar systems (globular clusters and faint inter-cluster light), and there is no indication of collapsed dark-matter structure analogous to baryonic stellar formation. The discovery of CDG-2 strengthens the broader phenomenological case that the dark sector lacks a first-order dissipation channel.

Scope of this comparison. We do not claim that CDG-2 uniquely confirms the SSM. Any dark-matter model with a non-dissipative dark sector — including the standard collisionless cold-dark-matter paradigm and many wave-like or fuzzy-dark-matter alternatives — is consistent with the existence of almost-dark galaxies. Furthermore, the 99.94–99.99% halo-mass fraction is inferred from GC-count scaling relations rather than direct kinematic measurement; the authors of [23] note that high-precision kinematic and spectroscopic follow-up is needed to confirm the dark-matter content directly. CDG-2 is therefore one observational data point consistent with the SSM phenomenology, not a discriminating test against alternative dark-matter models. The discovery of further almost-dark galaxies, particularly with direct kinematic confirmation of

their halo masses, would extend this class of observation and tighten the constraint on the morphological predictions of any non-dissipative dark-matter framework.

5.5 Comparison with existing dark matter models

The octahedral-void defect predicted here shares qualitative features with several composite-dark-matter scenarios in the literature. GeV-scale strongly-coupled hidden-sector models [16, 17] produce dark baryons whose mass arises from confinement dynamics and whose direct-detection signatures are suppressed relative to weakly-interacting massive particles. Strongly-interacting massive particle (SIMP) models [18] predict sub-GeV to GeV-scale dark particles with non-trivial self-interaction. The SSM prediction lies in a similar parameter region (1.7 GeV mass, suppressed electromagnetic coupling) but differs in origin: the existence and structural properties of the dark matter candidate are derived from FCC vacuum geometry rather than from cosmological freeze-out, asymmetric mechanisms, or hidden-sector gauge dynamics.

The closest phenomenological analog in the existing literature is the *sexaquark* [27, 28]: a hypothesized neutral, flavor-singlet, scalar bound state of $uuddss$ quarks with baryon number $B = 2$ and strangeness $S = -2$, proposed by Farrar as a dark matter candidate. The sexaquark and the SSM $K=6$ octahedral defect agree on several non-trivial phenomenological features. Both are GeV-scale particles with mass $\sim 2m_p$ (sexaquark stability requires $m_S \lesssim 2054$ MeV, with cosmological relic-abundance fits favoring 1.5–1.8 GeV [28]; the SSM prediction $m_{\text{DM}} = 1.719$ GeV lies inside this preferred range). Both involve a six-fold structure (sexaquark: six quarks; SSM defect: six bounding vertices). Both are electromagnetically neutral with suppressed hadronic couplings (sexaquark: flavor-singlet decoupling from pions; SSM: absence of square plaquettes in the bonded subgraph $K_{2,2,2}$). Both are consistent with the diffuse, non-collisional halo phenomenology required by direct-detection nulls and self-interaction constraints.

Three theoretical features distinguish the two models. First, the sexaquark carries $B = 2$ and $S = -2$ and is therefore strictly distinct from its antiparticle ($\bar{S}$, with $B = -2$, $S = +2$); the SSM defect, by contrast, is its own antiparticle by the inversion symmetry of the regular octahedron (Section 3.5). The Majorana-vs-Dirac character is empirically distinguishable in indirect-detection signatures (annihilation $\chi\chi$ vs. $\chi\bar{\chi}$) and in cosmological asymmetry channels. Second, the sexaquark mass is a free phenomenological parameter that lattice QCD cannot yet predict to the precision required for stability; the SSM mass follows from a closed combinatorial expansion on $K_{2,2,2}$ with no fitted parameters. Third, the sexaquark is a strongly-interacting QCD bound state (a flavor-singlet hadron), whereas the SSM defect carries no $\text{SU}(3)$ color charge (Section 3.4) and does not participate in strong interactions. The two frameworks are therefore not in direct competition: a definitive sexaquark detection would not falsify the SSM defect, and vice versa, though discriminating direct-detection, indirect-detection, and accelerator signatures can be designed.

6 Discussion

6.1 Falsifiable predictions

The framework makes the following predictions that could in principle be falsified by observation:

1. **Mass scale.** The framework predicts a single dark matter species with mass $m_{\text{DM}} = 1.719$ GeV, derived from the closed inclusion-exclusion expansion on $K_{2,2,2}$ (Eq. (6)) and the proton mass alone, with no cosmological input. Definitive identification of dark matter at a different mass scale, with the structural properties predicted here, would falsify the framework.
2. **Suppressed first-order EM coupling.** The octahedral defect has suppressed single-photon coupling at first order. Observation of first-order DM–photon coupling at the

level expected for charged or magnetic-moment-bearing dark matter would falsify the framework.

3. **Absence of standard SU(3) color.** The octahedral defect does not participate in QCD interactions. Observation of dark-matter hadronization or quark-binding signatures would falsify the framework.
4. **Single dark-matter species.** The framework predicts a single dark-matter species at the mass scale ~ 1.7 GeV. Definitive observation of multiple dark-matter species at distinctly different mass scales would require extension of the framework.
5. **Self-conjugate (Majorana-type) dark matter.** The octahedral defect is its own antiparticle. Observation of a Dirac-type dark sector with distinct particle/antiparticle, or asymmetric dark matter with a baryogenesis-like origin, would falsify the framework.
6. **Collisionless halos and absence of dark stellar collapse.** The framework predicts that dark matter cannot radiatively cool at first order and therefore forms diffuse non-collisional halos. Observation of dense compact dark-matter objects analogous to stars or planets, or evidence of strong dark-sector radiative cooling producing baryon-like collapse, would falsify the framework. The recent identification of CDG-2 [23] (see Section 5.4) is a positive observation consistent with this prediction.

6.2 Open calculations

The most important open calculations are, in order of priority:

1. *Forward derivation of $\Omega_{\text{DM}}/\Omega_{\text{b}}$.* The cosmological abundance ratio observed by Planck (5.36 ± 0.04 [1]) is not derived in the present paper. A first-principles SSM computation requires modeling the $K=4 \rightarrow K=12$ phase transition dynamics, the relative formation rates of octahedral and tetrahedral defects, the early-universe history of matter–antimatter annihilation in the tetrahedral sector, and $\chi\chi \rightarrow \gamma\gamma$ pair-annihilation in the octahedral sector. We identify this as the principal remaining open problem. The forward mass prediction $m_{\text{DM}} = 1.719$ GeV is independent of this calculation.
2. *QEC dual of the PIE expansion.* The structural counting $C_p = 1836$ for the proton has an independent realization as a fault-tolerant verification cost on a CSS code on the FCC lattice [3]. Constructing the analogous multi-cluster QEC dual of $C_{\text{DM}} = 3364$ for the octahedral defect would provide an independent check on the forward derivation. This is technically nontrivial because the octahedral defect’s bounding vertices span multiple coordination clusters (Section 4.5).
3. *Self-interaction cross-section.* A quantitative SSM calculation of the strain-field overlap between two octahedral defects would produce a self-interaction cross-section that can be compared to the Bullet Cluster bound and to the inferred small-scale-structure properties of dark-matter halos.
4. *Halo formation phenomenology.* The free-streaming length of a ~ 1.7 GeV particle decoupling at the FCC crystallization epoch determines whether the framework is consistent with cold-dark-matter structure formation or requires a warm/intermediate dark-matter treatment.

7 Conclusions

The Selection-Stitch Model identifies baryonic matter with $K=4$ remnants trapped in the tetrahedral interstitial voids of the FCC vacuum lattice. The same lattice contains a second inter-

stitial site — the octahedral void — that admits an analogous $K=6$ trapped remnant under the same kinematic rules. We have shown that this companion defect, identified with dark matter, has the structural properties expected of cold dark matter: suppressed first-order electromagnetic coupling (the bonding graph $K_{2,2,2}$ is bipartite-deficient at the relevant 4-cycle level and the bounding polyhedron has zero square faces), no $SU(3)$ color charge in the form generated for baryons (the skew-edge pair count $c_{\text{skew}}^{(O)} = 30$ does not factor as a three-color representation), a self-conjugate (Majorana-type) character due to the inversion symmetry of the regular octahedron, and suppressed first-order radiative cooling, which accounts for the observed baryonic–dark-matter clustering asymmetry.

The dark matter mass is predicted directly via the closed inclusion-exclusion expansion on the $K_{2,2,2}$ bonding graph, terminating exactly at third order by the 6-vertex constraint of the octahedron:

$$C_{\text{DM}} = 25 \cdot 144 - 30 \cdot 10 + 8 \cdot 8 = 3364, \quad m_{\text{DM}} = \frac{C_{\text{DM}}}{C_p} \times m_p = 1.719 \text{ GeV}.$$

This is a forward prediction using only the bonded structure of the defect, the structural-counting framework of Ref. [2], and the proton mass. No cosmological observation is consumed in the derivation. The recently reported 1.5–1.6 GeV gamma-ray line in three AGN [4] agrees with this prediction within 1–2.1 σ of each individual source measurement (joint fit at ~ 1.55 GeV, $\sim 11\%$ below the predicted rest-frame mass; the residual is consistent with gravitational redshift of inner-spike annihilation in the host SMBH environment). The cross-section structure required by the line ($\langle \sigma v \rangle_{\gamma\gamma} \approx 2 \times 10^{-28} \text{ cm}^3/\text{s}$, loop-suppressed) is consistent with the SSM prediction of suppressed first-order EM coupling. We caution that the Fermi-LAT signal remains at ~ 2.8 –4.1 σ in individual sources and requires confirmation.

The cosmological abundance ratio $\Omega_{\text{DM}}/\Omega_b$ is not derived in this paper. A first-principles SSM derivation requires modeling the $K=4 \rightarrow K=12$ phase transition dynamics, the relative formation rates of octahedral and tetrahedral defects, and the early-universe history of pair-annihilation in both the tetrahedral sector (matter–antimatter annihilation) and the octahedral sector ($\chi\chi \rightarrow \gamma\gamma$). We identify this as the principal remaining open problem.

The qualitative predictions of the framework — absence of first-order EM coupling, absence of $SU(3)$ color, self-conjugate Majorana character, and suppressed radiative cooling — follow from the structural symmetry of the $K_{2,2,2}$ bonding graph and the bounding octahedron, and are independent of the verification-cost calculation. The recent identification of the dark-matter-dominated almost-dark galaxy CDG-2 [23] provides observational support for the collisionless diffuse-halo prediction.

Data Availability

The geometric quantities used in this paper (FCC void radii, void counts per unit cell, edge graph $K_{2,2,2}$ for the octahedron, skew-pair count $c_{\text{skew}}^{(O)} = 30$ verified by three independent methods) are standard crystallographic and combinatorial data. The Planck cosmological parameters are from [1]. The Fermi-LAT line measurements are from [4]. The full enumeration of the 30 pairwise and 8 triple overlaps used in Eq. (6) is given in the Appendix; a self-contained reference implementation that reproduces every integer in the forward prediction $C_{\text{DM}} = 3364$ is available at https://github.com/raghu91302/ssmtheory/blob/main/verify_C_DM.py, and a companion script that verifies the orbit decomposition, body-diagonal labeling, and inversion pairing of the Embedding Uniformity Lemma (Section 4.4) is available at https://github.com/raghu91302/ssmtheory/blob/main/verify_uniformity.py. The interactive 3D visualization referenced in the introduction is hosted at https://raghu91302.github.io/ssmtheory/oct_void_3D.html.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix: Enumeration Tables for Pairwise and Triple Overlaps

This appendix provides the explicit enumeration of the 30 pairwise and 8 triple first-shell intersections that determine $K_{\text{pairwise}}^{(O)} = 10$ and $K_{\text{triple}}^{(O)} = 8$ in Eq. (6). The setup follows Section 4.3 exactly: the octahedral void is centered at $(1, 0, 0)$, the 12 FCC nearest-neighbor displacements are $\{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}$, and the first-shell neighborhood of an edge is $\mathcal{N}(e) = \mathcal{N}_1(v_a) \cup \mathcal{N}_1(v_b)$ with $|\mathcal{N}(e)| = 20$.

Vertex labels. We label the six bounding vertices of the octahedral void as in Table 2.

Label	Coordinates
A	$(2, 0, 0)$
B	$(0, 0, 0)$
C	$(1, +1, 0)$
D	$(1, -1, 0)$
E	$(1, 0, +1)$
F	$(1, 0, -1)$

Table 2: The six bounding vertices of the octahedral void at $(1, 0, 0)$. The three antipodal non-bonded pairs are (A, B) , (C, D) , (E, F) .

The 12 edges of $K_{2,2,2}$ are then $\{AC, AD, AE, AF, BC, BD, BE, BF, CE, CF, DE, DF\}$ (every vertex-pair from distinct antipodal classes). The 3 antipodal non-bonds AB, CD, EF are at distance $L\sqrt{2}$ and do not contribute to $K_{2,2,2}$.

All 30 pairwise intersections. Table 3 lists every skew pair of $K_{2,2,2}$ together with its first-shell intersection size. All 30 entries equal 10; no pair gives any other value. The uniformity is a consequence of the O_h symmetry of the octahedral void acting transitively on skew pairs.

$(AC, BD) : 10$	$(AD, BE) : 10$	$(AE, CF) : 10$
$(AC, BE) : 10$	$(AD, BF) : 10$	$(AE, DF) : 10$
$(AC, BF) : 10$	$(AD, CE) : 10$	$(AF, BC) : 10$
$(AC, DE) : 10$	$(AD, CF) : 10$	$(AF, BD) : 10$
$(AC, DF) : 10$	$(AE, BC) : 10$	$(AF, BE) : 10$
$(AD, BC) : 10$	$(AE, BD) : 10$	$(AF, CE) : 10$
$(AE, BF) : 10$	$(BC, DE) : 10$	$(AF, DE) : 10$
$(BC, DF) : 10$	$(BE, CF) : 10$	$(BD, CE) : 10$
$(BE, DF) : 10$	$(BF, CE) : 10$	$(BD, CF) : 10$
$(BF, DE) : 10$	$(CE, DF) : 10$	$(CF, DE) : 10$

Table 3: All 30 skew-edge pairs of $K_{2,2,2}$ with their first-shell intersection sizes $|\mathcal{N}(e_i) \cap \mathcal{N}(e_j)|$. Every entry is 10. The summary statistic $K_{\text{pairwise}}^{(O)} = 10$ used in Eq. (6) is the common value of this list, not an average over a non-uniform distribution.

All 8 perfect matchings and their triple intersections. Table 4 lists every perfect matching (3-matching) of $K_{2,2,2}$ together with the explicit 8 elements of its triple first-shell intersection. The decomposition into 6 bounding-vertex nodes plus 2 matching-specific bulk nodes is shown directly.

#	Matching	Bounding	Other	Other-node coordinates
M1	$\{AC, BE, DF\}$	6	2	$(0, +1, -1), (2, -1, +1)$
M2	$\{AC, BF, DE\}$	6	2	$(0, +1, +1), (2, -1, -1)$
M3	$\{AD, BE, CF\}$	6	2	$(0, -1, -1), (2, +1, +1)$
M4	$\{AD, BF, CE\}$	6	2	$(0, -1, +1), (2, +1, -1)$
M5	$\{AE, BC, DF\}$	6	2	$(0, -1, +1), (2, +1, -1)$
M6	$\{AE, BD, CF\}$	6	2	$(0, +1, +1), (2, -1, -1)$
M7	$\{AF, BC, DE\}$	6	2	$(0, -1, -1), (2, +1, +1)$
M8	$\{AF, BD, CE\}$	6	2	$(0, +1, -1), (2, -1, +1)$

Table 4: All 8 perfect matchings of $K_{2,2,2}$ with their first-shell triple intersections decomposed as $6 + 2$. The “Bounding” column counts the 6 bounding vertices A, B, C, D, E, F (common to every matching by construction), and the “Other” column counts the 2 matching-specific bulk nodes (listed explicitly in the last column). Every triple intersection has size 8, so $K_{\text{triple}}^{(O)} = 8$ uniformly.

Structure of the matching-specific nodes. The 2 matching-specific bulk nodes for each matching sit at displacement $(\pm 1, \pm 1, \pm 1)$ from the void center $(1, 0, 0)$, on the four body diagonals of the unit cube whose face centers are the bounding octahedral vertices. The 8 matchings group into 4 pairs $(M1 \leftrightarrow M8), (M2 \leftrightarrow M6), (M3 \leftrightarrow M7), (M4 \leftrightarrow M5)$, with the two matchings in each pair sharing the same 2-node companion set. The 4 distinct companion sets are in bijection with the 4 body diagonals of that cube. The two nodes within each companion set are inversion-related through the void center, reflecting the centrosymmetry $O_h \ni -I$ that also underlies the self-conjugate character of the defect (Section 3.5).

Assembly. Combining Tables 3 and 4 with $N_O = 25$ and the per-node disruption count $K^2 = 144$ reproduces Eq. (6) by direct substitution:

$$C_{\text{DM}} = 25 \times 144 - 30 \times 10 + 8 \times 8 = 3600 - 300 + 64 = 3364, \quad (12)$$

and $m_{\text{DM}} = (3364/1836) \times m_p = 1.7191 \text{ GeV}$ using the CODATA proton mass $m_p = 938.272 \text{ MeV}$ [8].